

NIKOLO-POLOMA, Russian Federation

WMO: 272520

Lat: 58.350N

Lon: 43.383E

Elev: 474

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-13.7	-24.9	1.4	-19.7	-18.4	2.0	-13.5	19.7	19.5	17.7	25.2	3.1	20	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	84.1	67.4	80.6	65.9	77.3	64.4	70.0	80.5	68.0	77.5	66.2	74.3	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.0	73.8	64.6	92.9	72.2	62.9	87.3	70.8	34.2	80.8	32.6	77.7	31.1	74.3	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.0	15.5	13.8	DB	-27.0	88.4	5.1	3.5	-30.6	90.9	-33.6	93.0	-36.4	95.0	-40.1	97.6
(5)				WB	-26.2	73.2	5.1	2.5	-29.9	75.0	-32.9	76.5	-35.8	77.9	-39.5	79.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	12.9	14.6	25.0	39.5	51.9	59.8	65.0	60.0	49.9	38.3	26.1	18.2
	DBStd	20.47	13.73	13.00	8.75	8.88	9.10	7.60	6.57	6.86	7.22	7.99	10.38	13.12
	HDD50	5509	1149	991	775	333	88	9	1	4	87	370	716	986
	HDD65	9803	1614	1411	1240	765	414	190	83	187	454	828	1166	1451
	CDD50	1343	0	0	0	18	147	305	467	313	85	8	0	0
	CDD65	160	0	0	0	0	8	36	84	31	1	0	0	0
	CDH74	1525	0	0	0	3	109	343	777	282	11	0	0	0
	CDH80	378	0	0	0	0	20	79	195	83	1	0	0	0
Wind	WSAvg	6.5	6.9	7.0	7.3	6.7	6.8	5.8	5.0	5.3	5.8	6.9	6.9	7.3
Precipitation	PrecAvg	24.8	1.5	1.1	1.2	1.4	1.9	3.4	2.8	2.8	2.2	2.6	1.9	1.6
	PrecMax	35.8	2.8	3.2	3.5	3.3	3.8	7.0	7.6	6.7	4.0	5.2	3.0	2.6
	PrecMin	19.0	0.6	0.2	0.1	0.6	0.8	0.9	0.1	1.0	0.7	0.9	0.4	0.6
	PrecStd	3.8	0.6	0.7	0.7	0.8	0.8	1.4	1.7	1.4	0.9	1.2	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.7	48.9	72.2	82.7	87.1	89.9	88.0	75.4	63.3	48.8	41.4
		MCWB	33.9	34.1	39.7	55.7	64.4	68.6	69.6	69.0	60.7	53.8	46.7	40.6
	2%	DB	33.5	34.4	42.2	65.8	76.7	81.9	84.9	80.4	70.0	56.5	43.4	36.9
		MCWB	32.7	32.9	35.9	51.5	60.6	66.2	68.0	67.0	59.6	50.3	41.9	35.8
	5%	DB	32.4	33.0	38.8	59.6	72.0	77.8	81.5	76.2	65.6	52.5	40.4	34.3
		MCWB	31.7	31.9	33.9	48.0	57.3	63.9	67.0	64.5	57.5	48.9	39.2	33.4
	10%	DB	30.7	31.6	36.3	54.5	67.8	74.0	78.4	72.2	61.8	49.3	37.8	32.8
		MCWB	29.9	30.7	33.1	45.1	54.4	62.2	66.0	62.5	55.4	46.9	36.8	32.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.3	34.8	40.8	56.2	66.5	71.0	72.9	71.5	64.4	55.1	46.9	41.0
		MCDB	34.7	35.6	46.7	69.7	79.1	82.5	82.1	83.9	70.1	60.5	48.4	41.4
	2%	WB	33.0	33.3	37.2	52.6	62.8	68.5	70.6	68.4	61.2	51.8	42.4	36.1
		MCDB	33.3	34.1	40.4	63.7	73.8	78.2	80.9	78.0	67.2	55.6	43.1	36.7
	5%	WB	31.8	32.3	35.2	49.5	59.7	66.0	68.7	66.2	58.6	49.3	39.3	33.7
		MCDB	32.3	33.1	37.5	57.8	68.1	74.6	78.5	73.2	64.2	51.7	40.1	34.1
	10%	WB	29.8	30.6	33.6	46.0	56.9	63.6	66.9	63.9	56.1	46.8	36.7	32.2
		MCDB	30.6	31.6	35.7	53.5	64.3	71.2	75.6	70.5	61.0	49.3	37.6	32.7
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.2	13.8	16.6	18.8	17.7	18.5	17.0	14.7	9.6	7.2	8.4
		MCDBR	6.9	8.5	15.4	25.8	25.2	23.7	23.7	22.7	21.0	16.1	8.7	6.2
		MCWBR	7.0	7.9	10.4	14.1	11.5	10.0	9.4	10.2	11.2	10.7	7.9	6.2
	5% WB	MCDBR	6.9	8.1	11.3	23.8	21.3	20.3	20.6	20.2	17.8	13.7	7.7	5.8
		MCWBR	7.1	7.9	8.4	13.9	11.0	9.6	9.5	9.9	10.0	10.1	7.7	5.9
Clear-Sky Solar Irradiance	taub		0.275	0.312	0.343	0.395	0.369	0.366	0.401	0.401	0.359	0.339	0.303	0.263
	taud		2.067	2.062	2.059	2.134	2.393	2.423	2.341	2.318	2.420	2.355	2.184	2.039
	Ebn at Noon		168	218	247	255	274	276	262	252	247	213	158	130
	Edh at Noon		17	28	37	41	34	33	36	34	26	21	15	12
All-Sky Solar Radiation	RadAvg		130	364	804	1219	1643	1722	1717	1303	763	326	120	72
	RadStd		23	89	100	154	130	142	158	104	100	60	19	15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.48	N/A	N/A	N/A	N/A	N/A	N/A	-423	N/A	N/A
(47)	Regional (6 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	-374	N/A	N/A

Nomenclature: See separate page